

A Counterexample to the Convergence of Three-Block ADMM with an Identity Third Constraint Block*

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Abstract

The alternating direction method of multipliers (ADMM), as a landmark algorithm, has attracted tremendous research attention and extensive practical applications over the past two decades [3]. It is well known that, although the two-block ADMM enjoys well-established theoretical convergence guarantees, its direct extension to the three-block case may fail to converge, as demonstrated by existing counterexamples [5]. However, to the best of our knowledge, the case in which the third constraint block is the identity remains unresolved: the existing literature gives neither a general convergence proof nor a counterexample for this subclass [5, 16, 17, 13]. In this paper, we aim to answer this open problem about ADMM. It is shown that the direct extension of three-block ADMM may nevertheless fail even when the first two blocks are strongly convex quadratics. We can formally construct an explicit counterexample with a strongly convex objective function and linear constraints satisfying the required conditions, such that when ADMM is directly applied to solve this problem, the resulting sequence fails to converge (forming a bounded nonconvergent sequence of minimal period 66). Furthermore, a small perturbation to the dual-variable update suffices to break the non-convergent cycle and yield a convergent method. The counterexample and further analysis were discovered in an AI-assisted workflow using Codex with the GPT-5.6 Sol model.

1 The Open Problem

Consider the linearly constrained three-block convex program

$$\begin{aligned} \min_{x_1, x_2, x_3} \quad & \theta_1(x_1) + \theta_2(x_2) + \theta_3(x_3) \\ \text{s.t.} \quad & A_1x_1 + A_2x_2 + A_3x_3 = b, \end{aligned} \tag{1.1}$$

where, for $i = 1, 2, 3$, $x_i \in \mathbb{R}^{n_i}$, $A_i \in \mathbb{R}^{m \times n_i}$, $b \in \mathbb{R}^m$, and $\theta_i : \mathbb{R}^{n_i} \rightarrow (-\infty, +\infty]$ is proper, closed, and convex. Block constraints may be absorbed into θ_i through indicator functions, and the solution set is assumed nonempty. Although two-block ADMM is convergent under standard assumptions [3, 6], its direct three-block extension can fail [5]. We study the subclass $n_3 = m$, $A_3 = I_m$.

*Code and certificates will be archived with the final version.

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This structure arises from the inequality-constrained problem

$$\begin{aligned} \min_{x,y} \quad & F(x) + G(y) \\ \text{s.t.} \quad & Ax + By \leq b, \end{aligned} \tag{1.2}$$

where $F : \mathbb{R}^{n_x} \rightarrow (-\infty, +\infty]$ and $G : \mathbb{R}^{n_y} \rightarrow (-\infty, +\infty]$ are proper, closed, and convex, $A \in \mathbb{R}^{m \times n_x}$, $B \in \mathbb{R}^{m \times n_y}$, and $b \in \mathbb{R}^m$. All vector inequalities are understood componentwise, and $\mathbb{R}_+^m := \{u \in \mathbb{R}^m : u \geq 0\}$. For a nonempty closed convex set C , let ι_C denote its indicator: $\iota_C(u) = 0$ for $u \in C$ and $\iota_C(u) = +\infty$ otherwise. Introducing a nonnegative slack variable gives

$$\begin{aligned} \min_{x,y,z} \quad & F(x) + G(y) + \iota_{\mathbb{R}_+^m}(z) \\ \text{s.t.} \quad & Ax + By + z = b. \end{aligned} \tag{1.3}$$

This is (1.1) with $(x_1, x_2, x_3) = (x, y, z)$ and $(A_1, A_2, A_3) = (A, B, I_m)$; the last subproblem is projection onto $\mathbb{R}_+^m$. We consider the direct slack-last sweep, rather than a grouped two-block reformulation [13].

For $\beta > 0$ and $\lambda \in \mathbb{R}^m$, we use the augmented Lagrangian

$$\mathcal{L}_\beta(x, y, z, \lambda) = F(x) + G(y) + \iota_{\mathbb{R}_+^m}(z) - \lambda^\top r + \frac{\beta}{2} \|r\|^2, \quad r = Ax + By + z - b \in \mathbb{R}^m. \tag{1.4}$$

Assuming that the subproblems attain solutions, direct three-block ADMM in the order $x \rightarrow y \rightarrow z \rightarrow \lambda$ is

$$x^{k+1} \in \arg \min_x \mathcal{L}_\beta(x, y^k, z^k, \lambda^k), \tag{1.5a}$$

$$y^{k+1} \in \arg \min_y \mathcal{L}_\beta(x^{k+1}, y, z^k, \lambda^k), \tag{1.5b}$$

$$z^{k+1} \in \arg \min_z \mathcal{L}_\beta(x^{k+1}, y^{k+1}, z, \lambda^k), \tag{1.5c}$$

$$\lambda^{k+1} = \lambda^k - \beta(Ax^{k+1} + By^{k+1} + z^{k+1} - b). \tag{1.5d}$$

We use the normal-cone convention

$$N_C(u) = \{v : \langle v, w - u \rangle \leq 0 \text{ for every } w \in C\}, \quad u \in C.$$

With the sign convention in (1.4), a KKT point satisfies

$$\begin{aligned} A^\top \lambda^* &\in \partial F(x^*), & B^\top \lambda^* &\in \partial G(y^*), \\ Ax^* + By^* + z^* &= b, & z^* &\in \mathbb{R}_+^m, & \lambda^* &\in N_{\mathbb{R}_+^m}(z^*). \end{aligned} \tag{1.6}$$

Thus $\lambda^* \leq 0$ and $(z^*)^\top \lambda^* = 0$.

Available multi-block convergence results impose additional regularity or parameter assumptions [18, 4, 16, 17], or modify the iteration by correction, proximal, or back-substitution steps [9, 14, 11, 10]. These guarantees do not follow from $A_3 = I_m$ alone. We first give an exact $A = B = I_2$ counterexample, then compare an independent certificate-construction route, and finally determine when a multiplier-step perturbation restores convergence.

2 The Counterexample

We state the rational QP, display its periodic orbit, and then certify that orbit by an exact piecewise-affine return map. The plot and branch spectra provide intuition; exact rational closure and projection signs provide the proof.

2.1 Counterexample data and main result

Set $A = B = I_2$, $\beta = 1$, and consider

$$\begin{aligned} \min_{x, y, z \in \mathbb{R}^2} \quad & \frac{1}{2}x^\top Q_1 x + \frac{1}{2}y^\top Q_2 y + \iota_{\mathbb{R}_+^2}(z) \\ \text{s.t.} \quad & x + y + z = \bar{b}, \end{aligned} \quad (2.1)$$

Define

$$M = (Q_1 + I_2)^{-1}, \quad N = (Q_2 + I_2)^{-1}.$$

Then (1.5) becomes

$$x^{k+1} = M(\bar{b} - y^k - z^k + \lambda^k), \quad (2.2a)$$

$$y^{k+1} = N(\bar{b} - x^{k+1} - z^k + \lambda^k), \quad (2.2b)$$

$$z^{k+1} = [\bar{b} - x^{k+1} - y^{k+1} + \lambda^k]_+, \quad (2.2c)$$

$$\lambda^{k+1} = \lambda^k - (x^{k+1} + y^{k+1} + z^{k+1} - \bar{b}). \quad (2.2d)$$

Set

$$\varepsilon = \frac{1}{1000}, \quad \mu = \frac{8957}{10000}, \quad \nu = \frac{999}{1000}, \quad d_1 = (-1, 20)^\top, \quad d_2 = (-1, 10)^\top, \quad (2.3)$$

and define

$$\begin{aligned} M &= \varepsilon I_2 + (\mu - \varepsilon) \frac{d_1 d_1^\top}{d_1^\top d_1}, \quad Q_1 = M^{-1} - I_2, \\ N &= \varepsilon I_2 + (\nu - \varepsilon) \frac{d_2 d_2^\top}{d_2^\top d_2}, \quad Q_2 = N^{-1} - I_2. \end{aligned} \quad (2.4)$$

To define the right-hand side, prescribe a KKT point. Choose

$$z^* = (0, 1)^\top, \quad \lambda^* = (-1, 0)^\top,$$

and set

$$x^* = Q_1^{-1} \lambda^*, \quad y^* = Q_2^{-1} \lambda^*, \quad \bar{b} = x^* + y^* + z^*. \quad (2.5)$$

By construction,

$$Q_1 x^* = \lambda^*, \quad Q_2 y^* = \lambda^*, \quad x^* + y^* + z^* = \bar{b}, \quad \lambda^* \in N_{\mathbb{R}_+^2}(z^*),$$

so $(x^*, y^*, z^*, \lambda^*)$ satisfies the KKT conditions (1.6). Every entry of $Q_1, Q_2, \bar{b}$ is rational. Moreover,

$$\sigma(Q_1) = \left\{ 999, \frac{1043}{8957} \right\}, \quad \sigma(Q_2) = \left\{ 999, \frac{1}{999} \right\}, \quad (2.6)$$

where $\sigma(Q)$ denotes the spectrum. Thus $Q_1, Q_2 \succ 0$. Strong convexity in (x, y) and the constraint determine a unique primal solution, while $Q_1 x^* = \lambda^*$ fixes the multiplier. Hence the KKT point is unique. All three ADMM subproblems also have unique minimizers, so the iteration is single-valued.

Theorem 2.1 (Exact rational period-66 counterexample). *For the rational problem (2.1)–(2.5), there is a rational initialization for which the sequence generated by the unmodified direct three-block ADMM (1.5) is bounded, non-KKT, and periodic with minimal period 66. Its strict projection-sign sequence is*

$$\mathcal{W} = (00)^2(01)^{64}, \quad (2.7)$$

where the i th bit is 1 if the corresponding projection input is positive and 0 if it is negative. All 132 projection inequalities are strict with a common margin greater than 10^{-3} . Therefore an identity third constraint block does not guarantee unconditional global convergence of direct three-block ADMM.

2.2 Orbit and geometric intuition

Figure 1 is a decimal visualization of the exact rational sequence in Theorem 2.1. The values $q^k = z^k + \lambda^k$ form a closed loop that does not contain the unique KKT value $q^* = (-1, 1)$.

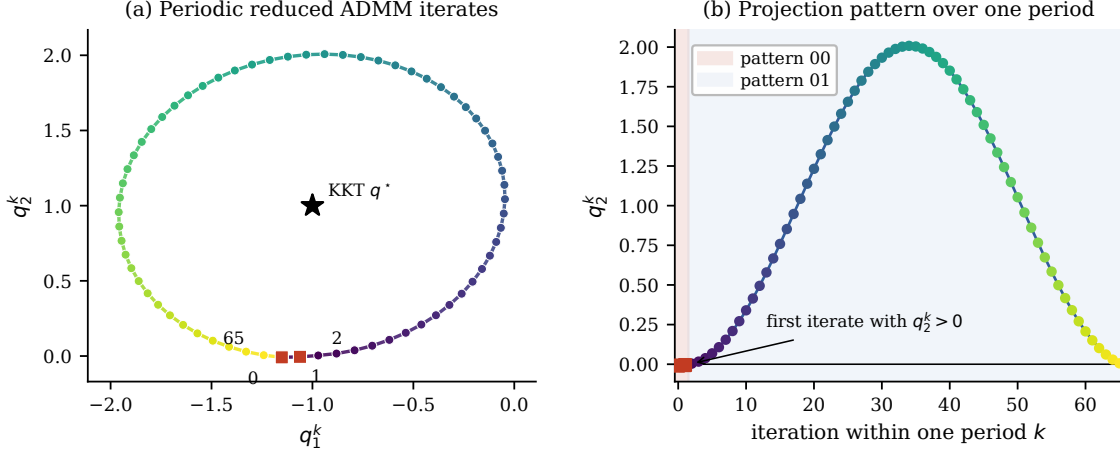


Figure 1: Decimal visualization of the exact rational period-66 sequence. Red squares mark the two 00 iterates; the remaining iterates have pattern 01. The KKT coordinate $q^* = (-1, 1)$ is not on the periodic sequence.

Numerically, the dominant 01-branch pair has modulus 1.00018 and angle $0.0951955 \approx 2\pi/66$, while $\rho(T_{00}) \approx 0.97058$. Thus 64 near-rotational 01 steps followed by two resetting 00 steps explain the word $(00)^2(01)^{64}$. This is only intuition; the proof follows.

2.3 Exact return map and period certificate

We now reduce the ADMM iteration to verify the periodic sequence in Theorem 2.1. For quadratic programs, local ADMM behavior and parameter selection can often be reduced to linear or affine iterations after active-set identification [2, 7, 15]. Here we instead compose an entire finite sign itinerary in exact rational arithmetic.

Define the orthant-projection input

$$q^{k+1} = \bar{b} - x^{k+1} - y^{k+1} + \lambda^k. \quad (2.8)$$

The z - and multiplier updates give

$$z^{k+1} = [q^{k+1}]_+, \quad \lambda^{k+1} = [q^{k+1}]_-, \quad q^{k+1} = z^{k+1} + \lambda^{k+1}. \quad (2.9)$$

For $a \in \mathbb{R}^2$, write $([a]_+)_i = \max\{a_i, 0\}$, $([a]_-)_i = \min\{a_i, 0\}$, and $(|a|)_i = |a_i|$ for $i = 1, 2$. Thus (2.9) shows that the single vector q^{k+1} uniquely determines both z^{k+1} and λ^{k+1} . Since $z^{k+1} = \Pi_{\mathbb{R}_+^2}(q^{k+1})$, the projection optimality condition gives

$$\lambda^{k+1} = q^{k+1} - z^{k+1} \in N_{\mathbb{R}_+^2}(z^{k+1}).$$

We index the recurrence after the projection-multiplier update and set

$$s^k = ((y^k)^\top, (q^k)^\top)^\top \in \mathbb{R}^4.$$

Since $z - \lambda = |q|$, the eliminated variables are recovered by

$$z = [q]_+, \quad \lambda = [q]_-, \quad x^+ = M(\bar{b} - y - z + \lambda) = M(\bar{b} - y - |q|). \quad (2.10)$$

Define $p := \bar{b} - x^+ - z + \lambda$ and $g := (I_2 - M)\bar{b}$. Substitution gives the reduced recurrence

$$p = My - (I_2 - M)|q| + g, \quad (2.11a)$$

$$y^+ = Np, \quad (2.11b)$$

$$q^+ = \bar{b} - x^+ - y^+ + \lambda = p + z - y^+ = (I_2 - N)p + [q]_+. \quad (2.11c)$$

Together with (2.10), this recurrence is equivalent to the original ADMM after the first projection-multiplier update.

For a strict projection pattern $D = \text{diag}(\mathbf{1}_{\{q_1 > 0\}}, \mathbf{1}_{\{q_2 > 0\}})$, one has $[q]_+ = Dq$ and $|q| = S_D q$, where $S_D = 2D - I_2$. Hence, on the corresponding polyhedral projection region, (2.11) is the affine update

$$s^+ = T_D s + h, \quad (2.12)$$

$$T_D = \begin{pmatrix} NM & -N(I_2 - M)S_D \\ (I_2 - N)M & D - (I_2 - N)(I_2 - M)S_D \end{pmatrix}, \quad h = \begin{pmatrix} Ng \\ (I_2 - N)g \end{pmatrix}.$$

Only two projection patterns are needed: $D_{00} = \text{diag}(0, 0)$ and $D_{01} = \text{diag}(0, 1)$. Write $T_{00} := T_{D_{00}}$ and $T_{01} := T_{D_{01}}$, and represent each affine update in homogeneous coordinates by

$$L_\omega = \begin{pmatrix} T_\omega & h \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{5 \times 5}, \quad \omega \in \{00, 01\}. \quad (2.13)$$

Composing the affine updates in the order prescribed by (2.7) gives

$$L_{01}^{64} L_{00}^2 = \begin{pmatrix} \mathcal{P} & a \\ 0 & 1 \end{pmatrix}. \quad (2.14)$$

Here $\mathcal{P} \in \mathbb{Q}^{4 \times 4}$ and $a \in \mathbb{Q}^4$. Exact rational elimination verifies $\det(I_4 - \mathcal{P}) \neq 0$. Hence $s^{66} = s^0$ has the unique rational solution

$$s^0 = (I_4 - \mathcal{P})^{-1} a \in \mathbb{Q}^4. \quad (2.15)$$

Thus the cycle is an affine fixed point selected by the nonzero offset a , which comes from $\bar{b}$, rather than a unit-eigenvalue orbit of $\mathcal{P}$ or an effect of linear objective terms. Prescribing a branch itinerary and then solving the fixed-point equation of its return map is related to constructive counterexample methodology [8]; the projection admissibility of every step must still be verified here.

It remains to verify that the fixed point (2.15) follows the prescribed projection branches. Exact rational iteration of (2.11) gives

$$q_1^k < 0 \quad (0 \leq k < 66), \quad q_2^k < 0 \quad (k = 0, 1), \quad q_2^k > 0 \quad (2 \leq k < 66). \quad (2.16)$$

More precisely, if $\delta_k = 0$ for $k = 0, 1$ and $\delta_k = 1$ otherwise, then

$$\min_{0 \leq k < 66} \min\{-q_1^k, (2\delta_k - 1)q_2^k\} > \frac{1}{1000}. \quad (2.17)$$

Hence every branch is admissible and the orthant projection realizes $(00)^2(01)^{64}$.

Equation (2.10) recovers the full ADMM sequence. Define

$$x^0 = M(\bar{b} - y^{65} - |q^{65}|).$$

Together with $z^0 = [q^0]_+$ and $\lambda^0 = [q^0]_-$, this gives $(x^0, y^0, z^0, \lambda^0) \in \mathbb{Q}^8$, and the step-65 update returns the full state. If the ADMM sequence had a proper subperiod, its projection-sign sequence would be a proper repetition. The only possible repetition factor is two, because the sequence has length 66 and contains exactly two occurrences of 00. Its two length-33 halves, however, contain different numbers of 00 symbols. The first return therefore occurs at step 66. The periodic sequence cannot contain the unique KKT point, which is a fixed point of this single-valued ADMM iteration. It is consequently non-KKT, nonconvergent, and bounded. This proves the theorem.

The strict projection margin and nonsingularity of the return equation also make the cycle persist under sufficiently small perturbations of the problem data; see Corollary A.1.

3 Comparison of GPT-5.6 Sol (Codex) and Kimi Code K3

After GPT-5.6 Sol through Codex produced the rational period-66 certificate in Theorem 2.1, we gave Kimi Code K3 the same problem statement, but not that candidate, in a separate run. It followed a different route and produced an $m = 3$, locally attracting non-KKT sequence of minimal period 23; Appendix B gives its exact rational certificate. Because $\hat{A}$ and $\hat{B}$ are nonsingular, Corollary B.2 shows that an invertible blockwise change of variables converts this example to an $[I_3, I_3, I_3]$ identity-slack QP while preserving every direct-ADMM iterate.

Table 1 compares the two realized routes descriptively. Because their problem construction, telemetry, and human interventions differ, it is not a controlled model benchmark.

Table 1: Descriptive comparison of the two realized certificate-construction routes.

Measure	GPT-5.6 Sol (Codex)	Kimi Code K3
Approximate run time	≈ 40 hours.	≈ 9 hours.
Recorded token volume	$\approx 1.704 \times 10^9$ total; 7.869×10^6 output.	$\approx 6.327 \times 10^7$ total; 4.823×10^5 output.
Certified result	Exact rational $m = 2$ identity-slack QP: bounded non-KKT sequence of minimal period 66 (Theorem 2.1).	Exact rational $m = 3$ identity-slack QP (after an invertible change of variables): a locally attracting non-KKT sequence of minimal period 23, with a certified open set of nonconvergent reduced initializations (Proposition B.1).
Computational route	Piecewise-affine reduction, structured active-set search, rationalization, and two exact replays.	Structured projector screening, KKT-compatible data construction, periodic-word detection, and exact rational certification.

4 Convergence under a dual-step perturbation

Keep the primal penalty $\beta > 0$ and replace only the multiplier update by

$$\lambda^{k+1} = \lambda^k - \tau(Ax^{k+1} + By^{k+1} + z^{k+1} - b), \quad \tau > 0. \quad (4.1)$$

The standard method corresponds to $\tau = \beta$. We first state a class-level result and its limitation, and then give sharper ranges for the period-66 instance.

4.1 Class-level convergence and its limitation

Proposition 4.1 (Problem-dependent small dual step). *In (1.3), suppose that $F : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$ and $G : \mathbb{R}^{n_y} \rightarrow \mathbb{R}$ are continuously differentiable, respectively μ_F - and μ_G -strongly convex, and have L_F - and L_G -Lipschitz gradients, where $\mu_F, \mu_G > 0$ and $L_F, L_G < \infty$. Assume also that $[A \ B]$ has full row rank. For every fixed $\beta > 0$, there exists a problem-dependent $\bar{\tau} > 0$ such that, for every $0 < \tau < \bar{\tau}$, the direct $x \rightarrow y \rightarrow z \rightarrow \lambda$ iteration with (4.1) converges R -linearly from every finite initialization to the unique KKT point.*

Proof. Let $u = (x, y, z)$, let $\bar{u}(\lambda)$ minimize the augmented Lagrangian at fixed λ , and set $d_\beta(\lambda) = \min_u \mathcal{L}_\beta(u; \lambda)$. Consider

$$V^k = \mathcal{L}_\beta(u^{k+1}; \lambda^k) - d_\beta(\lambda^k) + d_\beta(\lambda^*) - d_\beta(\lambda^k),$$

and write $S_k = \|u^{k+1} - u^k\|^2$ and $R_k = \|\nabla d_\beta(\lambda^k)\|^2$. Global smoothness, strong convexity, exact block minimization, and the global primal and dual error bounds give finite positive constants γ, C, C_p, C_d , depending only on the problem data and β , such that

$$V^k - V^{k-1} \leq -(\gamma - \tau C)S_k - \tau R_k, \quad V^k \leq C_p S_k + C_d R_k. \quad (4.2)$$

Hence $0 < \tau < \bar{\tau} := \gamma/C$ implies

$$V^k \leq \rho V^{k-1}, \quad \rho = (1 + \min\{(\gamma - \tau C)/C_p, \tau/C_d\})^{-1} < 1.$$

The iterate error is bounded by a constant multiple of V^k , which gives R -linear convergence. The full derivation and the initialization convention are given in Appendix A.1. $\square$

The threshold in Proposition 4.1 cannot be chosen independently of the problem; the obstruction already occurs in the strongly convex quadratic subclass. With $\vartheta = \tau/\beta$, the following result gives a quantitative obstruction.

Theorem 4.2 (No problem-independent positive relative-step interval). *For every*

$$0 < \vartheta \leq \frac{80}{119},$$

there is an $m = 3$ slack-last QP with $Q_1, Q_2 \succ 0$ and $[A \ B]$ of full row rank, together with a finite feasible initialization, for which direct $x \rightarrow y \rightarrow z \rightarrow \lambda$ ADMM with relative multiplier step ϑ generates a bounded nonconvergent sequence. Hence no $\vartheta_0 > 0$ depending only on the algorithm can guarantee convergence for the entire class.

Proof. Fix ϑ . Appendices A.3 and A.4 construct a continuous one-parameter branch matrix $U(\alpha) := U_D(r, \alpha)$, where $D = \text{diag}(1, 0, 0)$ and $r = \sqrt{7\vartheta/(80 - 7\vartheta)}$. The branch is unstable at the degenerate endpoint $\alpha = 0$, while the other endpoint $\alpha = \sqrt{2} - 1$ is strictly stable. Continuity gives an interior α_c with $\rho(T(\alpha_c)) = 1$; at this parameter the associated problem is already strongly convex and $[A \ B]$ has full row rank. Uniqueness of the KKT point excludes the eigenvalue $+1$, so the remaining unit-circle mode generates a bounded nonconvergent error sequence. Finally, the strict projection margin keeps every sufficiently small perturbation on the same affine branch; the appendix also constructs a feasible initialization and verifies boundedness of the full ADMM orbit. $\square$

When the value of $\bar{\tau}$ is unavailable, the dual step can instead be chosen by a backtracking acceptance test that evaluates one additional exact trial primal sweep.

Corollary 4.3 (General adaptive dual step with backtracking). *Under the assumptions of Proposition 4.1, write $u = (x, y, z)$, $E = [A, B, I_m]$, and set*

$$\kappa_p := \frac{1 + \max\{L_F, L_G\} + \beta\|E\|^2}{\lambda_{\min}(\text{diag}(\mu_F I, \mu_G I, 0) + \beta E^\top E)},$$

$$\text{PG}_\lambda(u) := u - \text{prox}_h(u - \nabla s_\lambda(u)), \quad B_E(v) := \|E\|^2 \kappa_p^2 \|v\|^2,$$

where $h(u) = \iota_{\mathbb{R}_+^m}(z)$ and s_λ is the smooth part of $\mathcal{L}_\beta(\cdot; \lambda)$, while $\mathcal{G}_\lambda$ denotes one exact $x \rightarrow y \rightarrow z$ Gauss–Seidel primal sweep at fixed multiplier. Fix $\chi, \delta \in (0, 1)$ and $\tau_{\max} > 0$. Algorithm 1 accepts a trial only when its $(\hat{v}, \hat{D})$ satisfy

$$\tau B_E(\hat{v}) \leq (1 - \chi)\hat{D}. \quad (4.3)$$

Under exact block solves and exact evaluation of the acceptance test, backtracking terminates finitely, the accepted steps have a positive lower bound independent of k , and $\{(u^{k+1}, \lambda^k)\}_{k \geq 1}$ converges globally R -linearly.

Proof. Appendix A.1 gives uniform constants $\gamma, \sigma > 0$: $B_E(\hat{v})$ controls the primal error, while $\hat{D} \geq \gamma\|\hat{u} - u^k\|^2$ and $\|\hat{v}\| \leq \sigma\|\hat{u} - u^k\|$. Consequently,

$$\tau_{\text{safe}, E} := \frac{(1 - \chi)\gamma}{\|E\|^2 \kappa_p^2 \sigma^2}, \quad \tau_k \geq \underline{\tau}_E := \min\{\tau_{\max}, \delta \tau_{\text{safe}, E}\} > 0,$$

and every $0 < \tau \leq \tau_{\text{safe}, E}$ passes the test. For the accepted trial, set $D_k := \hat{D}$. The variable-step potential and cost-to-go estimates in Appendix A.2 give

$$V^k - V^{k-1} \leq -\chi D_k - \tau_k \|\nabla d_\beta(\lambda^k)\|^2 \leq -\eta V^k, \quad \eta := \min\left\{\frac{\chi\gamma}{C_p}, \frac{\underline{\tau}_E}{C_d}\right\} > 0,$$

where C_p, C_d are independent of the step sequence. Thus $V^k \leq (1 + \eta)^{-1} V^{k-1}$, and the appendix error bounds yield global R -linear convergence. $\square$

Algorithm 1 Backtracking dual-step ADMM with an exact trial primal sweep

Require: $u^{\text{init}}, \lambda^0; \chi, \delta \in (0, 1), \tau_{\max} > 0$

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1:  $u^1 \leftarrow \mathcal{G}_{\lambda^0}(u^{\text{init}})$  ▷ initial primal sweep
2: for  $k = 1, 2, \dots$  do
3:    $r^k \leftarrow Eu^k - b$ 
4:   for  $j = 0, 1, 2, \dots$  do
5:      $\tau \leftarrow \tau_{\max} \delta^j$ 
6:      $\hat{\lambda} \leftarrow \lambda^{k-1} - \tau r^k$  ▷ trial dual iterate
7:      $\hat{u} \leftarrow \mathcal{G}_{\hat{\lambda}}(u^k)$  ▷ exact trial primal sweep
8:      $\hat{v} \leftarrow \text{PG}_{\hat{\lambda}}(u^k)$ 
9:      $\hat{D} \leftarrow \mathcal{L}_{\beta}(u^k; \hat{\lambda}) - \mathcal{L}_{\beta}(\hat{u}; \hat{\lambda})$ 
10:    if  $\tau B_E(\hat{v}) \leq (1 - \chi) \hat{D}$  then
11:       $(\tau_k, \lambda^k, u^{k+1}) \leftarrow (\tau, \hat{\lambda}, \hat{u})$  ▷ accept the trial iterate
12:      exit the inner loop
13:    else
14:      retain the current iterate pair  $(u^k, \lambda^{k-1})$ 
15:    end if
16:  end for
17: end for

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Quadratic weighted refinement. If

$$F(x) = \frac{1}{2}x^\top Q_1 x + c_1^\top x, \quad G(y) = \frac{1}{2}y^\top Q_2 y + c_2^\top y, \quad Q_1, Q_2 \succ 0,$$

set

$$K = \text{diag}(Q_1, Q_2, 0) + \beta E^\top E, \quad B_K(v) := \beta^{-1} v^\top (K^{-1} + 2I + K)v.$$

The appendix proves that $B_K(v)$ is also an upper bound for $\|E(u - \bar{u}(\lambda))\|^2$. Thus one may replace (4.3) by

$$\tau B_K(\hat{v}) \leq (1 - \chi) \hat{D}, \tag{4.4}$$

with the same finite-backtracking and convergence conclusions. Equivalently, replace B_E by B_K in the acceptance test of Algorithm 1. This refinement relies on the constant-Hessian structure of the quadratic objectives; it does not extend to general nonlinear objectives merely by substituting their current Hessians for the Q_i .

4.2 Sharper ranges for the period-66 instance

For the rational QP (2.1), set $\beta = 1$ and use the full state $w = (y, z, \lambda) \in \mathbb{R}^6$. On a strict projection branch with mask D , one complete update is affine:

$$w^+ = T_D(\tau)w + a_D(\tau), \quad T_D(\tau) = T_D^{(0)} + \tau T_D^{(1)}. \tag{4.5}$$

Here τ enters only the multiplier block row. The explicit block matrices are given in Appendix A.7. At the unique KKT point, $q^* = (-1, 1)$, so the local branch is D_{01} . Figure 2 separates the three scopes below.

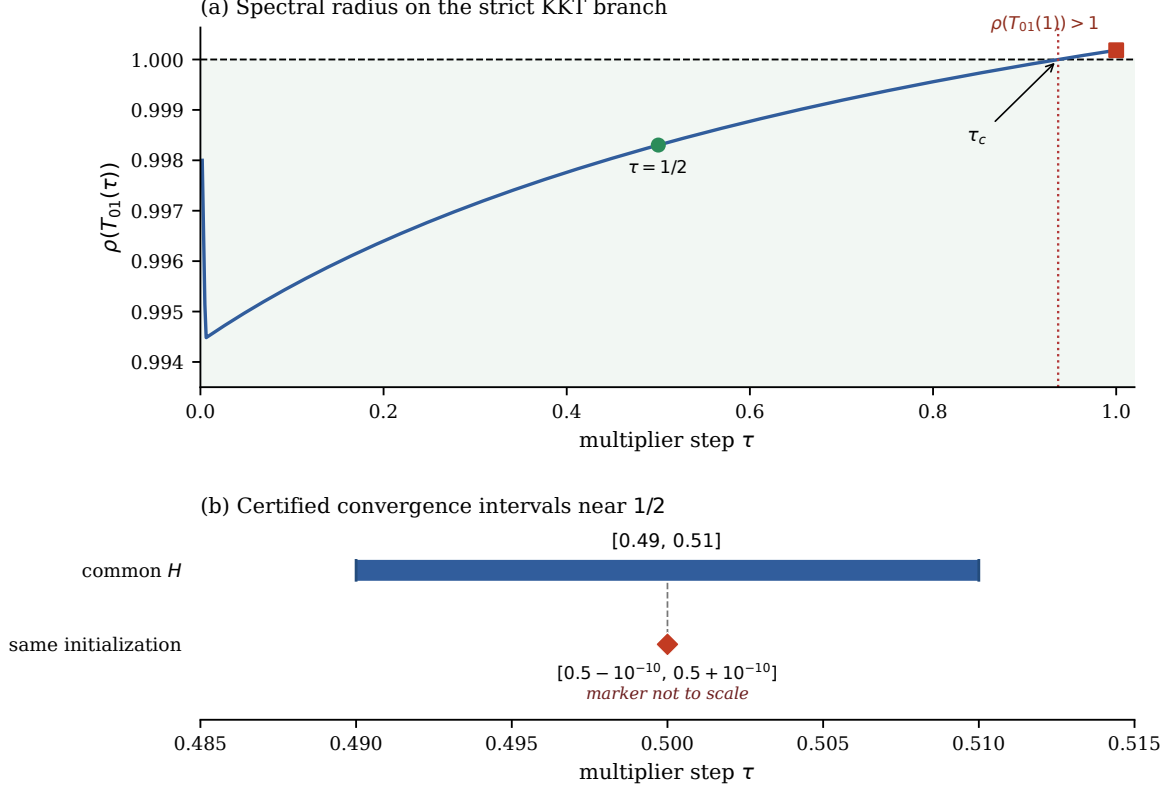


Figure 2: Certified dual-step ranges for the fixed rational instance: pointwise local attraction on the strict D_{01} branch, a uniform common-Lyapunov interval, and convergence of the initialization in Theorem 2.1.

Theorem 4.4 (Dual step length for the period-66 instance). *For the rational QP (2.1) and (4.1):*

1. *one rational Lyapunov matrix and one projection-safe neighborhood give uniform local linear convergence to the unique KKT point for*

$$\frac{49}{100} \leq \tau \leq \frac{51}{100};$$

2. *the initialization of Theorem 2.1 converges to that KKT point whenever*

$$\left| \tau - \frac{1}{2} \right| \leq 10^{-10};$$

3. *for each fixed $0 < \tau < 1$, the KKT point is locally linearly attracting for the full projected iteration if and only if $0 < \tau < \tau_c$, where*

$$0.9366061114 < \tau_c < 0.9366061115.$$

Equivalently, $T_{01}(\tau)$ is Schur stable exactly on this interval. The neighborhood and contraction metric may depend on τ .

Proof. At $\tau = 1/2$, an exact rational Lyapunov equation produces a common metric. Exact end-point tests and a quadratic chord identity extend contraction to $[0.49, 0.51]$; strict complementarity

supplies a projection-safe neighborhood. For the specified period-66 initialization, an exact 232-step interval enclosure for $|\tau - 1/2| \leq 10^{-10}$ enters that neighborhood. Finally, exact Schur recursion and a Sturm count identify the unique boundary τ_c ; inside the strict D_{01} branch, the resulting Lyapunov metric and the Banach fixed-point theorem give local linear convergence. The exact derivation, polynomial, and verification commands are in Appendices A.8 and A.9. $\square$

The intervals in the first two assertions are sufficient, not maximal. The first and third assertions are local KKT results; the second concerns only the initialization in Theorem 2.1. No fixed-instance arbitrary-initialization claim is made beyond Proposition 4.1.

5 Conclusion

The identity third constraint block does not guarantee global convergence of direct three-block ADMM. The two-dimensional period-66 certificate gives a bounded non-KKT orbit; the three-dimensional period-23 certificate further shows that such an orbit can be locally attracting on an open set of initializations.

Whether convergence is restored depends on the problem data. For globally smooth strongly convex objectives and full-row-rank $[A \ B]$, sufficiently small multiplier steps and the associated backtracking rule give global R-linear convergence, but no positive relative-step interval is uniform over the whole class. The fixed period-66 instance has a wider exact local stability range. Exact global ranges for a given problem, finite precision, and inexact block solves remain open.

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A Proofs and certificates for the period-66 and dual-step results

A.1 Proof of Proposition 4.1

Proof. Write $u = (x, y, z)$ and $E = [A, B, I_m]$. Decompose the augmented Lagrangian as

$$\mathcal{L}_\beta(u; \lambda) = s_\lambda(u) + h(u),$$

where

$$s_\lambda(u) := F(x) + G(y) - \lambda^\top (Eu - b) + \frac{\beta}{2} \|Eu - b\|^2, \quad h(u) := \iota_{\mathbb{R}_+^m}(z).$$

Set

$$d_\beta(\lambda) = \min_u \mathcal{L}_\beta(u; \lambda).$$

We first verify uniqueness of the KKT point. Full row rank of $[A \ B]$ implies strict feasibility: for any $z > 0$, the equation $[A \ B](x, y) = b - z$ is solvable. Strong convexity gives a unique primal solution. In the present multiplier convention, the unaugmented negative dual objective is

$$\phi_0(\lambda) = F^*(A^\top \lambda) + G^*(B^\top \lambda) - b^\top \lambda + \iota_{\mathbb{R}^m_-}(\lambda).$$

Standard conjugacy results imply that an L -smooth convex function has a $1/L$ -strongly convex conjugate [1]. Hence F^* and G^* are $1/L_F$ - and $1/L_G$ -strongly convex, respectively. Consequently,

$$m_0 = \lambda_{\min} \left(\frac{AA^\top}{L_F} + \frac{BB^\top}{L_G} \right) > 0,$$

where strict positivity follows from the full-row-rank assumption. The multiplier is therefore unique.

We next establish the global primal error bound at fixed multiplier. Set

$$K_\mu = \text{diag}(\mu_F I, \mu_G I, 0) + \beta E^\top E, \quad \mu_p = \lambda_{\min}(K_\mu) > 0,$$

$$L_p = \max\{L_F, L_G\} + \beta \|E\|^2.$$

For all u, v ,

$$\begin{aligned} \langle \nabla s_\lambda(u) - \nabla s_\lambda(v), u - v \rangle &\geq (u - v)^\top K_\mu (u - v) \geq \mu_p \|u - v\|^2, \\ \|\nabla s_\lambda(u) - \nabla s_\lambda(v)\| &\leq L_p \|u - v\|. \end{aligned}$$

Let $\bar{u}(\lambda)$ denote the unique minimizer of $\mathcal{L}_\beta(\cdot; \lambda)$. The proximal-gradient residual mapping

$$\text{PG}_\lambda(u) := u - \text{prox}_h(u - \nabla s_\lambda(u))$$

satisfies, by proximal optimality and monotonicity of ∂h ,

$$\|u - \bar{u}(\lambda)\| \leq \kappa_p \|\text{PG}_\lambda(u)\|, \quad \kappa_p = \frac{1 + L_p}{\mu_p},$$

globally, with a constant independent of λ .

Exact $x \rightarrow y \rightarrow z$ block minimization gives

$$\gamma = \frac{1}{2} \min \left\{ \lambda_{\min}(\mu_F I + \beta A^\top A), \lambda_{\min}(\mu_G I + \beta B^\top B), \beta \right\} > 0$$

and

$$\mathcal{L}_\beta(u^k; \lambda^k) - \mathcal{L}_\beta(u^{k+1}; \lambda^k) \geq \gamma \|u^{k+1} - u^k\|^2.$$

Let

$$\begin{aligned} a_x &= L_F + \beta \|A^\top A\|, \quad a_y = L_G + \beta \|B^\top B\|, \\ \sigma^2 &= a_x^2 + \beta^2 \|B^\top A\|^2 + a_y^2 + \beta^2 \|A\|^2 + \beta^2 \|B\|^2 + (\beta + 2)^2. \end{aligned}$$

For $\Delta x = x^k - x^{k+1}$, $\Delta y = y^k - y^{k+1}$, and $\Delta z = z^k - z^{k+1}$, write the three blocks of $\text{PG}_{\lambda^k}(u^k)$ as $(\text{PG}_{x,k}, \text{PG}_{y,k}, \text{PG}_{z,k})$. The block optimality conditions and nonexpansiveness of the orthant projection then give

$$\begin{aligned} \|\text{PG}_{x,k}\| &\leq a_x \|\Delta x\|, \\ \|\text{PG}_{y,k}\| &\leq \beta \|B^\top A\| \|\Delta x\| + a_y \|\Delta y\|, \\ \|\text{PG}_{z,k}\| &\leq \beta \|A\| \|\Delta x\| + \beta \|B\| \|\Delta y\| + (\beta + 2) \|\Delta z\|. \end{aligned}$$

Consequently,

$$\|\text{PG}_{\lambda^k}(u^k)\| \leq \sigma \|u^{k+1} - u^k\|.$$

For the x, y blocks this uses only global Lipschitz continuity of $\nabla F, \nabla G$; for z , it compares the projection fixed-point identities at the current point and the block minimizer.

By the standard Moreau-envelope calculus [1], the negative augmented dual $-d_\beta$ is the Moreau envelope of ϕ_0 , and therefore has strong-convexity modulus

$$m_d = \frac{m_0}{1 + \beta m_0} > 0.$$

It follows that

$$\|\lambda - \lambda^*\| \leq m_d^{-1} \|\nabla d_\beta(\lambda)\|, \quad \nabla d_\beta(\lambda) = b - E\bar{u}(\lambda).$$

Define

$$\begin{aligned} \Delta_p^k &= \mathcal{L}_\beta(u^{k+1}; \lambda^k) - d_\beta(\lambda^k), \\ \Delta_d^k &= d_\beta(\lambda^*) - d_\beta(\lambda^k), \\ V^k &= \Delta_p^k + \Delta_d^k. \end{aligned}$$

The gap calculation of Hong and Luo [12, Theorem 3.1, Lemma 3.1, and (3.16)–(3.17)], in the present sign convention, yields

$$V^k - V^{k-1} \leq -a \|u^{k+1} - u^k\|^2 - \tau \|\nabla d_\beta(\lambda^k)\|^2, \quad a = \gamma - \tau \|E\|^2 \kappa_p^2 \sigma^2.$$

Thus $a > 0$ whenever

$$0 < \tau < \frac{\gamma}{\|E\|^2 \kappa_p^2 \sigma^2}.$$

The cost-to-go estimate on the noncompact orthant can be verified directly. Set $\Delta x = x^k - x^{k+1}$, $\Delta y = y^k - y^{k+1}$, and $\Delta z = z^k - z^{k+1}$, and define

$$H_{\text{GS}} = \beta \begin{pmatrix} 0 & A^\top B & A^\top \\ 0 & 0 & B^\top \\ 0 & 0 & 0 \end{pmatrix}, \quad C_p = \frac{\|H_{\text{GS}}\|^2}{2\mu_p}.$$

The three exact block optimality conditions, including the normal-cone condition for z , give

$$w^{k+1} = \begin{pmatrix} -\beta A^\top (B\Delta y + \Delta z) \\ -\beta B^\top \Delta z \\ 0 \end{pmatrix} \in \partial_u \mathcal{L}_\beta(u^{k+1}; \lambda^k).$$

Uniform μ_p -strong convexity at fixed multiplier therefore yields

$$\Delta_p^k \leq \frac{\|w^{k+1}\|^2}{2\mu_p} \leq C_p \|u^{k+1} - u^k\|^2.$$

Moreover, $q_\beta = -d_\beta$ is m_d -strongly convex and $1/\beta$ -smooth, so one may take

$$C_d = \frac{1}{2\beta m_d^2}, \quad \Delta_d^k \leq C_d \|\nabla d_\beta(\lambda^k)\|^2.$$

These constants are independent of k, λ^k , and the dual step. Consequently,

$$\eta = \min\{a/C_p, \tau/C_d\} > 0, \quad \rho = (1 + \eta)^{-1} \in (0, 1)$$

imply

$$V^k - V^{k-1} \leq -\eta V^k, \quad V^k \leq \rho V^{k-1}.$$

Finally, strong convexity and Lipschitz continuity of the fixed-multiplier minimizer give

$$\begin{aligned} \|u^{k+1} - \bar{u}(\lambda^k)\|^2 &\leq 2\Delta_p^k/\mu_p, \\ \|\lambda^k - \lambda^*\|^2 &\leq 2\Delta_d^k/m_d, \\ \|\bar{u}(\lambda^k) - u^*\| &\leq (\|E\|/\mu_p)\|\lambda^k - \lambda^*\|. \end{aligned}$$

Hence there is a constant $C > 0$ for which

$$\|(u^{k+1}, \lambda^k) - (u^*, \lambda^*)\| \leq C\rho^{k/2}.$$

If the initial z^0 is infeasible, the argument starts after its first projection onto $\mathbb{R}_+^m$. The compact-polyhedral assumption in the original Hong–Luo theorem is not invoked directly here: the global smooth strongly convex structure above replaces the error bounds for which that assumption is used. $\square$

A.2 Proof details for the adaptive dual-step acceptance test

Fix an arbitrary candidate τ , and abbreviate its trial values by

$$\lambda = \hat{\lambda}, \quad u = u^k, \quad u^+ = \hat{u}, \quad D = \mathcal{L}_\beta(u; \lambda) - \mathcal{L}_\beta(u^+; \lambda).$$

Let

$$\begin{aligned} e &= u - \bar{u}(\lambda), \\ v &= \text{PG}_\lambda(u). \end{aligned}$$

The global primal error bound in Proposition 4.1 directly gives

$$\|Ee\|^2 \leq \|E\|^2\|e\|^2 \leq \|E\|^2\kappa_p^2\|v\|^2 = B_E(v). \quad (\text{A.1})$$

The trial decrease can be evaluated from the block optimality conditions, without subtracting two nearby augmented-Lagrangian values. After the initial primal sweep, $z, z^+ \in \mathbb{R}_+^m$. If $\Delta x = x - x^+$, $\Delta y = y - y^+$, and

$$q = b - Ax^+ - By^+ + \lambda/\beta, \quad z^+ = [q]_+,$$

and if

$$D_F(a, c) = F(a) - F(c) - \langle \nabla F(c), a - c \rangle$$

with D_G defined analogously, cancellation of the first-order terms in the block optimality conditions gives

$$\begin{aligned} D &= D_F(x, x^+) + \frac{\beta}{2}\|A\Delta x\|^2 \\ &\quad + D_G(y, y^+) + \frac{\beta}{2}\|B\Delta y\|^2 \\ &\quad + \frac{\beta}{2}(\|z - q\|^2 - \|z^+ - q\|^2). \end{aligned} \quad (\text{A.2})$$

Exact block minimization gives uniform $\gamma, \sigma > 0$ such that

$$D \geq \gamma\|u^+ - u\|^2, \quad \|v\| \leq \sigma\|u^+ - u\|.$$

Consequently every

$$0 < \tau \leq \tau_{\text{safe}, E} := \frac{(1 - \chi)\gamma}{\|E\|^2\kappa_p^2\sigma^2}$$

satisfies (4.3). A geometric search starting from $\tau_{\max}$ therefore terminates and accepts

$$\tau_k \geq \underline{\tau}_E := \min\{\tau_{\max}, \delta\tau_{\text{safe}, E}\} > 0.$$

For the accepted sequence, set $D_k = \mathcal{L}_\beta(u^k; \lambda^k) - \mathcal{L}_\beta(u^{k+1}; \lambda^k)$. With $\bar{u}^k = \arg \min_u \mathcal{L}_\beta(u; \lambda^k)$, $r^k = Eu^k - b$, $\bar{r}^k = E\bar{u}^k - b$, and $d_k = d_\beta(\lambda^k)$, direct expansion gives

$$V^k - V^{k-1} = -D_k + \tau_k \|r^k\|^2 - 2(d_k - d_{k-1}).$$

Concavity of d_β and $\lambda^{k-1} - \lambda^k = \tau_k r^k$ imply

$$d_k - d_{k-1} \geq \tau_k \langle \bar{r}^k, r^k \rangle.$$

Completing the square yields

$$V^k - V^{k-1} \leq \tau_k \|E(u^k - \bar{u}^k)\|^2 - D_k - \tau_k \|\nabla d_\beta(\lambda^k)\|^2. \quad (\text{A.3})$$

Equations (4.3) and (A.1) imply

$$V^k - V^{k-1} \leq -\chi D_k - \tau_k \|\nabla d_\beta(\lambda^k)\|^2.$$

The proof of Proposition 4.1 explicitly constructs step-sequence-independent constants $C_p, C_d > 0$ such that

$$\Delta_p^k \leq (C_p/\gamma) D_k, \quad \Delta_d^k \leq C_d \|\nabla d_\beta(\lambda^k)\|^2.$$

Using $\tau_k \geq \underline{\tau}_E$ gives

$$V^k \leq (1 + \eta)^{-1} V^{k-1}, \quad \eta = \min \left\{ \frac{\chi\gamma}{C_p}, \frac{\underline{\tau}_E}{C_d} \right\} > 0. \quad (\text{A.4})$$

This closes the proof for the general smooth strongly convex class.

Quadratic weighted refinement. If F, G are the strongly convex quadratics stated in the main text, set

$$K = \text{diag}(Q_1, Q_2, 0) + \beta E^\top E, \quad p = \text{prox}_h(u - \nabla s_\lambda(u)) = u - v.$$

Proximal optimality and monotonicity of ∂h give

$$e^\top K e + \|v\|^2 \leq e^\top (I + K) v.$$

Cauchy–Schwarz in the K -metric, together with $K \succeq \beta E^\top E$, yields

$$\begin{aligned} e^\top (I + K) v &\leq \sqrt{e^\top K e} \sqrt{v^\top (K^{-1} + 2I + K) v}, \\ \|Ee\|^2 &\leq \frac{1}{\beta} e^\top K e \leq \frac{1}{\beta} v^\top (K^{-1} + 2I + K) v = B_K(v). \end{aligned} \quad (\text{A.5})$$

This step uses the quadratic identity $\nabla s_\lambda(u) - \nabla s_\lambda(\bar{u}) = Ke$.

The majorant can be evaluated without forming K^{-1} . For $v = (v_x, v_y, v_z)$,

$$\begin{aligned} v^\top K^{-1} v &= (v_x - A^\top v_z)^\top Q_1^{-1} (v_x - A^\top v_z) \\ &\quad + (v_y - B^\top v_z)^\top Q_2^{-1} (v_y - B^\top v_z) + \beta^{-1} \|v_z\|^2, \end{aligned} \quad (\text{A.6})$$

$$v^\top K v = v_x^\top Q_1 v_x + v_y^\top Q_2 v_y + \beta \|Av_x + Bv_y + v_z\|^2. \quad (\text{A.7})$$

Thus two pre-factorized solves with Q_1 and Q_2 suffice. The two Bregman terms in (A.2) reduce to $\frac{1}{2}\|\Delta x\|_{Q_1}^2$ and $\frac{1}{2}\|\Delta y\|_{Q_2}^2$.

Set

$$c_K = \frac{1}{\beta} \max_{\xi \in \sigma(K)} (\xi + \xi^{-1} + 2).$$

Since $B_K(v) \leq c_K \|v\|^2$, every

$$0 < \tau \leq \tau_{\text{safe},K} := \frac{(1-\chi)\gamma}{c_K \sigma^2}$$

satisfies (4.4), and geometric backtracking accepts

$$\tau_k \geq \underline{\tau}_K := \min\{\tau_{\text{max}}, \delta\tau_{\text{safe},K}\} > 0.$$

Replacing (4.3)–(A.1) by (4.4)–(A.5) in (A.3) gives the same contraction, with $\underline{\tau}_K$ in place of $\underline{\tau}_E$.

The trial-point evaluation order is essential: the trial residual $\hat{v}$ and decrease $\hat{D}$ must use the same trial dual iterate. If a candidate step is rejected, both trial iterates must be discarded, and the next candidate must be constructed from the same (u^k, λ^{k-1}) . The theorem assumes exact solves of the block subproblems and exact evaluation of the acceptance test; the strict margin needed for a finite-precision acceptance decision requires separate analysis.

A.3 Proof of Theorem 4.2

Proof. It suffices to take $\beta = 1$. For a prescribed $0 < \vartheta \leq 80/119$, set

$$r = \sqrt{\frac{7\vartheta}{80-7\vartheta}} \in (0, 1/4], \quad u = (1, r, 0)^\top, \quad v = (1, 3r/5, 4r/5)^\top,$$

and define

$$P_A = \frac{uu^\top}{1+r^2}, \quad P_B = \frac{vv^\top}{1+r^2}.$$

For $\alpha \in [0, \sqrt{2}-1]$, let

$$\begin{aligned} s &= \frac{2\alpha}{1+\alpha^2}, & t &= \frac{1-\alpha^2}{1+\alpha^2}, \\ A_\alpha &= tP_A + s(I - P_A), & Q_{1,\alpha} &= s^2P_A + t^2(I - P_A), \\ B_\alpha &= tP_B + s(I - P_B), & Q_{2,\alpha} &= s^2P_B + t^2(I - P_B). \end{aligned}$$

Then $Q_{1,\alpha} + A_\alpha^\top A_\alpha = Q_{2,\alpha} + B_\alpha^\top B_\alpha = I$. For $\alpha > 0$, both Hessians are positive definite and A_α, B_α are nonsingular. Take

$$x^* = y^* = 0, \quad z^* = (1, 0, 0)^\top, \quad \lambda^* = (0, -1, -1)^\top,$$

and set

$$b = z^*, \quad c_1 = A_\alpha^\top \lambda^*, \quad c_2 = B_\alpha^\top \lambda^*.$$

This is the unique strictly complementary KKT point.

Fix $D = \text{diag}(1, 0, 0)$. The exact Schur recursion in Appendix A.4 proves

$$\rho(U_D(r, 0)) > 1, \quad \vartheta = \frac{80r^2}{7(1+r^2)}.$$

At $\alpha = \sqrt{2} - 1$,

$$A_\alpha = B_\alpha = I/\sqrt{2}, \quad Q_{1,\alpha} = Q_{2,\alpha} = I/2,$$

and the coordinates decouple. The scalar characteristic polynomials for a positive-slack and a zero-slack coordinate are

$$\frac{1}{4}(2\zeta - 1)^2(\zeta + \vartheta - 1), \quad \frac{\zeta}{4}[4\zeta^2 + (3\vartheta - 5)\zeta + 1 - \vartheta].$$

Write $p(\zeta) = 4\zeta^2 + (3\vartheta - 5)\zeta + 1 - \vartheta$. The Jury inequalities for this quadratic factor are

$$4 > |1 - \vartheta|, \quad p(1) = 2\vartheta > 0, \quad p(-1) = 10 - 4\vartheta > 0,$$

so this endpoint is strictly Schur stable throughout the stated interval.

By continuity, some $\alpha_c \in (0, \sqrt{2} - 1)$ satisfies $\rho(U_D(r, \alpha_c)) = 1$. The corresponding problem remains strongly convex and full row rank. The branch cannot have eigenvalue $+1$: a sufficiently small perturbation along its eigenvector would remain in the strict affine branch and create a second ADMM fixed point, contradicting uniqueness of the KKT point.

The unit-circle eigenvalue is therefore -1 or a nonreal conjugate pair. Choose a genuine eigenmode: a real eigenvector for -1 , or the real invariant subspace spanned by the real and imaginary parts of a complex eigenvector. For a nonzero e in that mode, $\{U_D(r, \alpha_c)^k e\}$ alternates or rotates. It is therefore bounded but nonconvergent. Scaling e below the strict KKT projection margin keeps every iterate in the same branch, so the branch orbit is a bounded nonconvergent raw ADMM orbit. Because the unit-circle eigenvalue is nonzero, the two inactive components of the z -part of e vanish; taking e still smaller preserves the positive active component and gives $z^0 \in \mathbb{R}_+^3$. Since A_{α_c} is nonsingular, set

$$x^0 = A_{\alpha_c}^{-1}(b - B_{\alpha_c}y^0 - z^0).$$

Then $A_{\alpha_c}x^0 + B_{\alpha_c}y^0 + z^0 = b$, so the initial state is fully feasible. The stored x^0 does not enter the right-hand side of the next x -subproblem and therefore does not alter the (y, z, λ) orbit above. Each subsequent x -update is an affine function of that bounded reduced state, so the full (x, y, z, λ) orbit is bounded as well. $\square$

A.4 Symbolic certificate for the universal-step obstruction

For $e = (\delta y, \delta z, \delta \lambda)$, $R_A = A_\alpha A_\alpha^\top$, and $R_B = B_\alpha B_\alpha^\top$, define

$$\begin{aligned} Y_y &= B_\alpha^\top R_A B_\alpha, & Y_z &= -B_\alpha^\top (I - R_A), & Y_\lambda &= B_\alpha^\top (I - R_A), \\ C_y &= (I - R_B) R_A B_\alpha, & C_z &= R_A - R_B R_A + R_B, & C_\lambda &= I - R_A - R_B + R_B R_A. \end{aligned}$$

On a fixed projection mask D , the full error matrix is

$$U_D(r, \alpha) = \begin{pmatrix} Y_y & Y_z & Y_\lambda \\ DC_y & DC_z & DC_\lambda \\ \vartheta(I - D)C_y & \vartheta(I - D)C_z & (1 - \vartheta)I + \vartheta(I - D)C_\lambda \end{pmatrix}.$$

At $\alpha = 0$, $D = \text{diag}(1, 0, 0)$, and $\vartheta = 80r^2/[7(1 + r^2)]$, remove the four zero roots and the root $1 - \vartheta$ from the characteristic polynomial. The first three exact Schur deltas of the remaining quartic satisfy

$$\Delta_0 > 0, \quad \Delta_1 > 0, \quad \Delta_2 < 0 \quad (0 < r \leq 1/4),$$

where, after positive factors are removed, the signs of Δ_1 and Δ_2 are determined by

$$\begin{aligned} -h_1(r), \quad h_1(r) &= 1205r^8 + 1977r^6 - 4643r^4 - 7385r^2 - 2450, \\ h_2(r), \quad h_2(r) &= 14530r^{10} + 57137r^8 - 131931r^6 \\ &\quad - 449323r^4 - 329735r^2 - 82950. \end{aligned}$$

For $0 < r \leq 1$,

$$h_1(r) \leq -1461r^4 - 7385r^2 - 2450 < 0,$$

and

$$h_2(r) \leq -60264r^6 - 449323r^4 - 329735r^2 - 82950 < 0.$$

Thus the signs are strict and a root lies outside the unit disk. The exact polynomial, factorization, and endpoint Jury checks are regenerated by

```
python experiments/analyze_identity_slack_universal_step_obstruction.py.
```

The corresponding JSON is under `outputs/universal_dual_step/`.

A.5 Exact replay for Theorem 2.1

From the period-66 certificate repository root, run

```
python python/verify_certificate_pair.py.
```

The command invokes separately implemented four- and six-dimensional rational replays and verifies strong convexity, the unique KKT point, every ADMM update, 66-step closure, all 132 strict projection inequalities, and minimality of the period.

A.6 Persistence under perturbations of the problem data

Corollary A.1 (Persistence under data perturbations). *Let $\mathbb{S}_{++}^2$ denote the cone of real 2×2 symmetric positive-definite matrices. With $A = B = I_2$, $\beta = 1$, and the update order $x \rightarrow y \rightarrow z \rightarrow \lambda$ fixed, there is an open neighborhood of $(Q_1, Q_2, \bar{b})$ in $\mathbb{S}_{++}^2 \times \mathbb{S}_{++}^2 \times \mathbb{R}^2$ such that each resulting problem has a non-KKT periodic ADMM sequence of minimal period 66, with the same projection-sign sequence and every projection inequality strict.*

Proof. For the fixed projection-sign sequence, $\mathcal{P}$ and a in (2.14) depend analytically on the problem data. The period system remains nonsingular near the displayed instance, so $(I_4 - \mathcal{P})^{-1}a$ and its 66 reduced iterates vary continuously. The signs in (2.17) have a positive common margin and remain strict under sufficiently small perturbations. The primitive word is unchanged. $\square$

A.7 Full-state branch map for the relaxed period-66 instance

Set $\beta = 1$,

$$M = (Q_1 + I_2)^{-1}, \quad N = (Q_2 + I_2)^{-1}, \quad I = I_2,$$

and use $w = (y, z, \lambda) \in \mathbb{R}^6$. Eliminating x^+ from the primal updates gives

$$C_y = (I - N)M, \quad C_z = N + (I - N)M, \quad C_\lambda = (I - N)(I - M), \quad d = C_\lambda \bar{b},$$

and

$$q^+ = C_y y + C_z z + C_\lambda \lambda + d.$$

For a strict projection mask

$$D = \text{diag}(\mathbf{1}_{\{q_1^+ > 0\}}, \mathbf{1}_{\{q_2^+ > 0\}}), \quad z^+ = Dq^+,$$

the identity $x^+ + y^+ + z^+ - \bar{b} = \lambda - q^+ + z^+$ yields

$$\lambda^+ = (1 - \tau)\lambda + \tau(I - D)q^+.$$

Thus the matrices in (4.5) are

$$T_D^{(0)} = \begin{pmatrix} NM & -N(I - M) & N(I - M) \\ DC_y & DC_z & DC_\lambda \\ 0 & 0 & I \end{pmatrix},$$

$$T_D^{(1)} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ (I - D)C_y & (I - D)C_z & (I - D)C_\lambda - I \end{pmatrix},$$

and

$$a_D(\tau) = \begin{pmatrix} N(I - M)\bar{b} \\ Dd \\ 0 \end{pmatrix} + \tau \begin{pmatrix} 0 \\ 0 \\ (I - D)d \end{pmatrix}.$$

This also shows directly that τ affects only the multiplier block row.

A.8 Proof of Theorem 4.4

Proof. Set $T_0 := T_{01}^{(0)}$, $T_1 := T_{01}^{(1)}$, and $T(\tau) = T_{01}(\tau) = T_0 + \tau T_1$. Define $\Phi_H(\tau) = H - T(\tau)^\top H T(\tau)$. If $\tau = \eta\tau_- + (1 - \eta)\tau_+$, direct expansion gives

$$\Phi_H(\tau) - \eta\Phi_H(\tau_-) - (1 - \eta)\Phi_H(\tau_+) = \eta(1 - \eta)(\tau_+ - \tau_-)^2 T_1^\top H T_1 \succeq 0. \quad (\text{A.8})$$

At $\tau = 1/2$, let $H \in \mathbb{Q}^{6 \times 6}$ solve

$$H - T_{01}(1/2)^\top H T_{01}(1/2) = I_6.$$

Exact Sylvester tests give $H \succ 0$, $\Phi_H(49/100) \succ 0$, and $\Phi_H(51/100) \succ 0$. Equation (A.8) therefore gives uniform contraction throughout $[49/100, 51/100]$. Strict complementarity places a sufficiently small closed H -ellipsoid around the KKT point entirely inside the D_{01} projection region, proving the first assertion.

For the initialization in Theorem 2.1, a 232-step rational componentwise enclosure is propagated uniformly on $|\tau - 1/2| \leq 10^{-10}$. Every projection sign remains strict, and the entire enclosure at step 232 lies in the preceding projection-safe ellipsoid. The first assertion then proves convergence.

For the third assertion, exact factorization of the D_{01} -branch characteristic polynomial, followed by a real Schur recursion and a Sturm root count, reduces the stability boundary to the unique root in $(0, 1)$ of p_{stab} in (A.9); exact endpoint evaluation gives the stated bracket. If $0 < \tau < \tau_c$, Schur stability gives the unique $H_\tau \succ 0$ satisfying

$$H_\tau - T_{01}(\tau)^\top H_\tau T_{01}(\tau) = I_6.$$

In the induced norm there is a $\kappa_\tau < 1$ such that

$$\|T_{01}(\tau)e\|_{H_\tau} \leq \kappa_\tau \|e\|_{H_\tau}.$$

Since $q^* = (-1, 1)$ is in the interior of the strict branch, a sufficiently small closed H_τ -ellipsoid is invariant under the full projected map. The Banach fixed-point theorem gives local linear convergence to the unique KKT point. Conversely, when $\tau_c \leq \tau < 1$, the branch spectral radius is at least one; because the KKT point is interior to this branch, it cannot be locally linearly attracting. $\square$

A.9 Verification of Theorem 4.4

The dual-step certificate is verified by

```
python python/certify_relaxed_multiplier_interval_theory.py.
```

It re-derives the full-state projection-region matrices from the original updates, checks the end-point Sylvester minors and chord identity, propagates the exact 232-step finite-entry enclosure, and performs the Schur–Sturm boundary test. The boundary polynomial is

$$\begin{aligned} p_{\text{stab}}(\tau) = & 111794210406295556649228900462157733493 \tau^3 \\ & + 23105776975281816108275814441284422085171521 \tau^2 \\ & - 244157339715898821440243649673959463071543521 \tau \\ & + 208410060660460340386576638889814578828638507. \end{aligned} \tag{A.9}$$

An exact Sturm count gives one root of p_{stab} in $(0, 1)$, and rational endpoint evaluation gives the bracket in Theorem 4.4.

B Exact certificate for the period-23 counterexample

This appendix records the exact $m = 3$ certificate obtained by the Kimi Code K3 route. It is a fixed-QP initialization result and does not assert robustness to perturbations of the problem data.

B.1 Reduced branch map

Record the state immediately after the z - and multiplier updates. With $\beta = 1$, set

$$t = z + \lambda, \quad v = (y, t) \in \mathbb{R}^6, \quad S_t = [0_{3 \times 3} \ I_3],$$

so that $t = S_t v$. The projection identity gives $z = [t]_+$ and $\lambda = [t]_-$; thus v determines (z, λ) and the next ADMM step.

Each strict projection branch is determined by

$$\sigma(t) := (\text{sgn } t_1, \text{sgn } t_2, \text{sgn } t_3) \in \{-, +\}^3.$$

On a fixed branch the projection is linear, and eliminating the x - and z -updates gives

$$\Phi_\sigma(v) = R_\sigma v + r_\sigma, \quad R_\sigma \in \mathbb{Q}^{6 \times 6}, \quad r_\sigma \in \mathbb{Q}^6.$$

For quadratic objectives

$$F(x) = \frac{1}{2} x^\top Q_x x + c_1^\top x, \quad G(y) = \frac{1}{2} y^\top Q_y y + c_2^\top y,$$

set

$$\begin{aligned} K_x &= (Q_x + A^\top A)^{-1}, \quad K_y = (Q_y + B^\top B)^{-1}, \quad H_x = A K_x A^\top, \quad H_y = B K_y B^\top, \\ D_\sigma &= \text{diag}(\mathbf{1}_{\{\sigma_i = +\}}), \quad J_\sigma = 2D_\sigma - I_3, \quad E_\sigma = I_3 - D_\sigma, \\ \xi &= K_x(A^\top b - c_1), \quad \eta = K_y(B^\top b - c_2 - B^\top A \xi). \end{aligned}$$

Direct substitution in the x -, y -, and projection–multiplier updates gives

$$R_\sigma = \begin{bmatrix} K_y B^\top H_x B & K_y B^\top (H_x - I_3) J_\sigma \\ (I_3 - H_y) H_x B & E_\sigma + ((I_3 - H_y) H_x + H_y) J_\sigma \end{bmatrix}, \quad r_\sigma = \begin{bmatrix} \eta \\ b - A \xi - B \eta \end{bmatrix}. \tag{B.1}$$

If the QP data are rational, then every entry of R_σ and r_σ is rational.

B.2 Rational counterexample and exact certificate

With $\beta = 1$, apply the reduced map above to the following rational instance:

$$\begin{aligned}\widehat{A} &= \begin{bmatrix} 5/36 & 1/60 & 25/97 \\ 1/60 & 5/77 & 5/91 \\ 25/97 & 5/91 & 11/12 \end{bmatrix}, & \widehat{B} &= \begin{bmatrix} 4/61 & 1/54 & 5/83 \\ 1/54 & 13/92 & 23/88 \\ 5/83 & 23/88 & 53/58 \end{bmatrix}, \\ \widehat{Q}_x &= \begin{bmatrix} 85/93 & -1/57 & -3/11 \\ -1/57 & 99/100 & -5/86 \\ -3/11 & -5/86 & 7/78 \end{bmatrix}, & \widehat{Q}_y &= \begin{bmatrix} 99/100 & -1/51 & -3/47 \\ -1/51 & 72/79 & -18/65 \\ -3/47 & -18/65 & 4/43 \end{bmatrix}, \\ \widehat{b} &= \begin{bmatrix} -2/17 \\ -1/14 \\ 56/73 \end{bmatrix}, & \widehat{c}_1 &= \begin{bmatrix} 33/98 \\ -33/25 \\ -24/55 \end{bmatrix}, & \widehat{c}_2 &= \begin{bmatrix} -31/99 \\ 19/36 \\ -11/20 \end{bmatrix}.\end{aligned}\tag{B.2}$$

The resulting strict projection-sign word is

$$\mathcal{W}_{23} = (+, -, +)^5 (-, +, +)^7 (-, -, +)^2 (-, -, -) (-, -, +)^8.\tag{B.3}$$

Take $\omega = \mathcal{W}_{23} = (\sigma_0, \dots, \sigma_{22})$. Write the composition of its first j branch maps as

$$\begin{aligned}\Phi_\omega^{(j)}(v) &= L_j v + d_j, & L_0 &= I_6, & d_0 &= 0, \\ L_{j+1} &= R_{\sigma_j} L_j, & d_{j+1} &= R_{\sigma_j} d_j + r_{\sigma_j}.\end{aligned}$$

The full 23-step return map is therefore

$$\begin{aligned}v^{23(\ell+1)} &= \Phi_{\text{per}}(v^{23\ell}), \\ \Phi_{\text{per}}(v) &:= \Phi_\omega^{(23)}(v) = M_{\text{per}} v + c_{\text{per}}, \\ M_{\text{per}} &:= L_{23} = R_{\sigma_{22}} \cdots R_{\sigma_0}, & c_{\text{per}} &:= d_{23}.\end{aligned}\tag{B.4}$$

Exact elimination shows that $I_6 - M_{\text{per}}$ is nonsingular, so define the phase-zero state by

$$\widehat{v}^0 := (I_6 - M_{\text{per}})^{-1} c_{\text{per}} \in \mathbb{Q}^6, \quad \widehat{v}^j := L_j \widehat{v}^0 + d_j \quad (j = 0, \dots, 22).$$

The complete exact rational initialization is recovered by

$$\begin{aligned}\widehat{t}^0 &= S_t \widehat{v}^0, & \widehat{z}^0 &= [\widehat{t}^0]_+, & \widehat{\lambda}^0 &= [\widehat{t}^0]_-, \\ \widehat{K}_x &:= (\widehat{Q}_x + \widehat{A}^\top \widehat{A})^{-1}, & \widehat{x}^0 &= \widehat{K}_x \left[\widehat{A}^\top (\widehat{b} - \widehat{B} \widehat{y}^{22} - \widehat{z}^{22} + \widehat{\lambda}^{22}) - \widehat{c}_1 \right].\end{aligned}$$

These equations define the exact rational initialization; the machine certificate stores its expanded numerators and denominators.

Proposition B.1 (Exact period-23 certificate for nearby initializations). *For the rational data (B.2), $\widehat{Q}_x, \widehat{Q}_y \succ 0$ and $\det(\widehat{A}) \det(\widehat{B}) \neq 0$. The direct iteration (1.5), with $\beta = 1$, applied to*

$$\begin{aligned}\min_{x, y, z \in \mathbb{R}^3} & \quad \frac{1}{2} x^\top \widehat{Q}_x x + \widehat{c}_1^\top x + \frac{1}{2} y^\top \widehat{Q}_y y + \widehat{c}_2^\top y + \iota_{\mathbb{R}_+^3}(z) \\ \text{s.t.} & \quad \widehat{A}x + \widehat{B}y + z = \widehat{b},\end{aligned}\tag{B.5}$$

has the following properties.

- (i) It has a unique primal–dual KKT point.

- (ii) *Direct ADMM has a non-KKT sequence of minimal period 23. Its strict projection-sign word is (B.3); its exponents denote consecutive repetitions and sum to 23. All $23 \times 3 = 69$ projection inputs are separated from zero:*

$$\min_{0 \leq k < 23} \min_{1 \leq i \leq 3} |\hat{t}_i^k| > \frac{1}{250}.$$

- (iii) *There is a rational matrix $P \succ 0$ for which*

$$\mathcal{E}_{\text{cert}} := \left\{ \hat{v}^0 + e : e^\top P e < \frac{1}{4000} \right\} \quad (\text{B.6})$$

is an open invariant set for the 23-step return map (B.4). Every $v^0 \in \mathcal{E}_{\text{cert}}$ follows the repeating pattern $\mathcal{W}_{23}$ and converges phasewise to the period-23 sequence.

Consequently, the reconstructed ADMM sequence is nonconvergent for every reduced initialization $v^0 \in \mathcal{E}_{\text{cert}}$.

Proof. For $\omega = \mathcal{W}_{23}$, the recursion above gives the exact rational return map

$$(M_{\text{per}}, c_{\text{per}}) = (L_{23}, d_{23}).$$

Exact rational elimination solves the fixed-point system for $\hat{v}^0$. Exact replay reconstructs every phase state and verifies the pattern, the strict margin, $\Phi_{\text{per}}(\hat{v}^0) = \hat{v}^0$, and that the 23 phase states are pairwise distinct. Hence the sequence has minimal period 23.

Positive definiteness of $\hat{Q}_x$ and $\hat{Q}_y$ gives a unique primal solution; feasibility determines z , and nonsingularity of $\hat{A}$ determines λ . Thus the KKT point is unique and is a fixed point of the single-valued ADMM map. It cannot belong to the nonconstant period-23 sequence, which is therefore non-KKT.

For nearby initializations, the matrix P in Appendix B.3 satisfies

$$P - M_{\text{per}}^\top P M_{\text{per}} \succ 0.$$

It follows that the error decreases after each complete block of 23 ADMM steps. In particular, every eigenvalue of M_{per} lies strictly inside the unit disk. The strict sign margin guarantees that every point in $\mathcal{E}_{\text{cert}}$ continues to use the same 23 affine branches. The return map sends this ellipsoid into itself, so induction gives the same projection pattern and phasewise convergence for all subsequent returns. The 23 limiting phase points are distinct; hence the full sequence does not converge to a single point. $\square$

Corollary B.2 (Identity-slack equivalence). *Set*

$$u = \hat{A}x, \quad w = \hat{B}y.$$

Then problem (B.5) is equivalent to

$$\begin{aligned} \min_{u, w, z \in \mathbb{R}^3} \quad & \frac{1}{2} u^\top \tilde{Q}_x u + \tilde{c}_1^\top u + \frac{1}{2} w^\top \tilde{Q}_y w + \tilde{c}_2^\top w + \iota_{\mathbb{R}_+^3}(z) \\ \text{s.t.} \quad & u + w + z = \hat{b}, \end{aligned}$$

where

$$\tilde{Q}_x = \hat{A}^{-\top} \hat{Q}_x \hat{A}^{-1}, \quad \tilde{Q}_y = \hat{B}^{-\top} \hat{Q}_y \hat{B}^{-1}, \quad \tilde{c}_1 = \hat{A}^{-\top} \hat{c}_1, \quad \tilde{c}_2 = \hat{B}^{-\top} \hat{c}_2.$$

The transformed data remain rational and $\tilde{Q}_x, \tilde{Q}_y \succ 0$. Thus Proposition B.1 also gives an exact rational, locally attracting period-23 counterexample for the $[I_3, I_3, I_3]$ identity-slack model.

Proof. Nonsingularity of $\hat{A}$ and $\hat{B}$ makes the change of variables bijective and preserves the constraint residual:

$$\hat{A}x + \hat{B}y + z - \hat{b} = u + w + z - \hat{b}.$$

The x - and y -subproblems are therefore invertible reparameterizations of the u - and w -subproblems, while the z - and multiplier updates are unchanged. Starting from $u^0 = \hat{A}x^0$ and $w^0 = \hat{B}y^0$, the two direct ADMM trajectories consequently correspond step by step. Positive definiteness is preserved by congruence, and inverses of nonsingular rational matrices remain rational. $\square$

B.3 Invariant ellipsoid and local attraction of the period-23 orbit

Use the partial-composition matrices L_j and the selector S_t defined in Appendix B.1. Set

$$P = \frac{1}{2} \begin{bmatrix} 2 & 0 & 0 & 0 & -1 & 1 \\ 0 & 2 & 1 & 0 & -2 & 2 \\ 0 & 1 & 6 & 2 & -7 & 7 \\ 0 & 0 & 2 & 3 & -3 & 3 \\ -1 & -2 & -7 & -3 & 16 & -13 \\ 1 & 2 & 7 & 3 & -13 & 14 \end{bmatrix}.$$

Its leading principal minors are

$$1, \quad 1, \quad \frac{11}{4}, \quad \frac{25}{8}, \quad \frac{333}{32}, \quad \frac{441}{32},$$

so $P \succ 0$. Exact Sylvester tests also give

$$P - M_{\text{per}}^\top P M_{\text{per}} \succ 0.$$

For the i th standard basis vector $\mathbf{e}_i \in \mathbb{R}^3$, define

$$a_{ji} := \mathbf{e}_i^\top S_t L_j \in \mathbb{Q}^{1 \times 6}.$$

Thus $a_{ji}e$ is the perturbation of the i th projection input at phase j caused by the initial-state error $e = v^0 - \hat{v}^0$. Define the corresponding sign-preserving radius by

$$\bar{r}^2 := \min_{\substack{0 \leq j < 23, 1 \leq i \leq 3 \\ a_{ji} \neq 0}} \frac{(\hat{t}_i^j)^2}{a_{ji} P^{-1} a_{ji}^\top}.$$

The exact certificate gives $\bar{r}^2 > 29/100000 > 1/4000$; the minimum occurs at phase 14 and coordinate t_3 . The P -norm Cauchy–Schwarz inequality therefore preserves all 69 projection signs for $e^\top P e < 1/4000$. Since $P - M_{\text{per}}^\top P M_{\text{per}} \succ 0$, the return map leaves $\mathcal{E}_{\text{cert}}$ invariant. Induction gives the same branch word and phasewise convergence to the non-KKT sequence.

The ellipsoid in (B.6) is sufficient but not claimed maximal. On the slice $\Delta y = 0$, $\Delta t_3 = 0$,

$$\frac{3}{2}(\Delta t_1)^2 - 3\Delta t_1 \Delta t_2 + 8(\Delta t_2)^2 < \frac{1}{4000}, \quad \Delta t_2 = 0 \implies |\Delta t_1| < \frac{1}{\sqrt{6000}}.$$

Every point in this interval is therefore a certified nonconvergent reduced initialization.

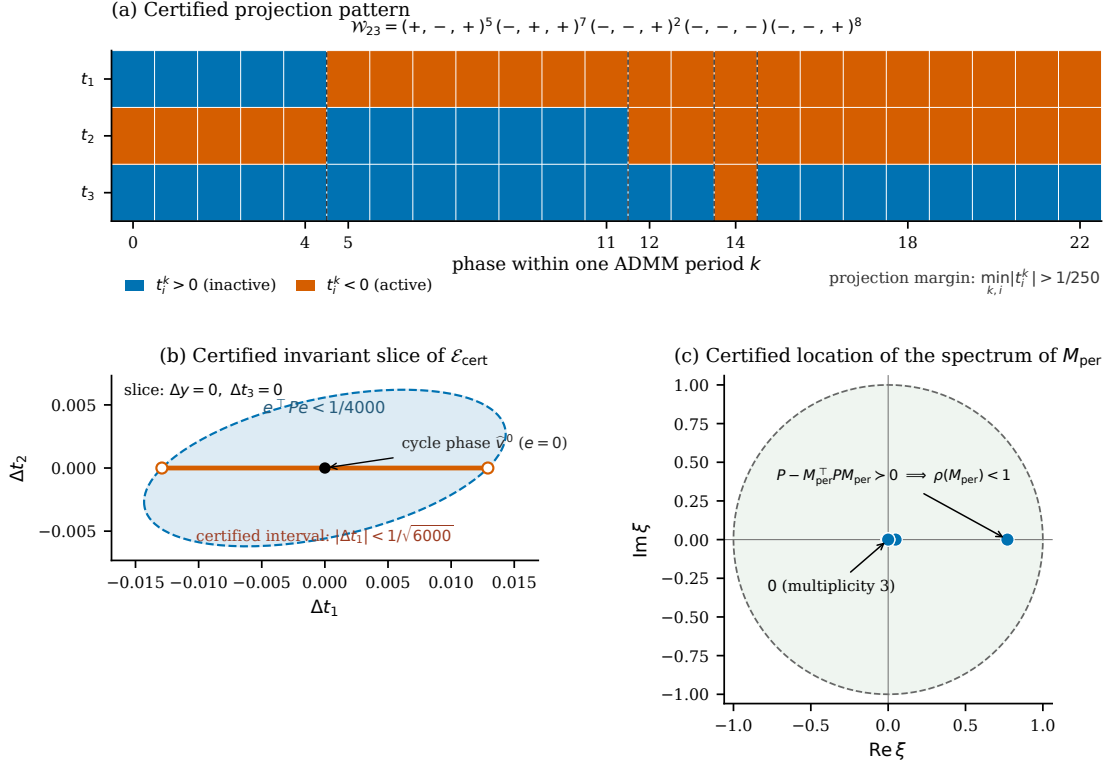


Figure 3: Local-attraction certificate for the period-23 orbit: (a) the projection word $\mathcal{W}_{23}$ and its strict sign margin; (b) a slice of the invariant ellipsoid and an explicit certified interval; (c) the return-matrix spectrum. The exact Lyapunov inequality certifies $\rho(M_{\text{per}}) < 1$; no claim is made outside the certified ellipsoid.

B.4 Reproduction and source data

From the public certificate repository root, running

```
python python/verify_period23_certificate.py
```

verifies positive definiteness, nonsingularity, the unique KKT point, 23-step closure, 23 distinct phase states, strict projection consistency with margin $> 1/250$, separation from the KKT point, and the exact Lyapunov inequality for the return matrix. The rational input, verifier, and generated certificate are provided in the accompanying repository.

C Comparison protocol

The Codex workflow included human interventions in problem specification, evidence requirements, and search priorities. The Kimi workflow began from a frozen workspace without Codex candidates, but was audited post hoc rather than preregistered.

Table 1 uses the first exact replayable certificate as the endpoint. Wall-clock time deduplicates completed-turn intervals; token totals include cached input, noncached input, and output. The two telemetry and tokenization pipelines differ, so these figures document the runs rather than rank their efficiency.